

Searching the recursion tree improves the lower bound on the Shannon capacity of C_7

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Abstract

The recent sequence of improvements to the lower bound on the Shannon capacity of the seven-cycle shares a common shape: a library of admissible combining rules is applied along a fixed recursion tree. The rules have been studied carefully; the tree has not. We show that optimising the tree, using no data beyond what is already published, gives

$$\Theta(C_7) \geq 3.258834362237710794\dots,$$

improving the bound $3.258832620353266309\dots$ of Tandon. The construction is K_{4a} applied to four copies of $S_{3b}(n_6, n_{11}, n_{11})$, where n_6 and n_{11} are Tandon’s own published cardinality vectors and S_{3b} , K_{4a} are rules of Buys, Polak and Zuiddam: the only change is that the third input is a second copy of n_{11} rather than n_8 . Among all 16 384 combinations of the three published blocks, the published choice is third by value.

We also record what does *not* help, since several natural avenues can be closed exhaustively rather than merely explored. The base gadget of Polak and Schrijver admits exactly eight private pairs and this is a structural ceiling; its neutral auxiliary part cannot exceed 322 over the entire automorphism orbit; nine and even ten private pairs do exist, on other 367-sets, but are incompatible with a full-sized auxiliary set, and the obstruction is quantitative – the region the auxiliary set must avoid grows by about 142 vertices with each extra pair; and the published twist is optimal over all 64 538 880 automorphisms. Finally we note two traps that internal consistency does not catch.

1 Introduction

Shannon’s zero-error capacity $\Theta(G)$ [1] is known for no odd cycle beyond C_5 , where Lovász’s bound [2] settles it. For C_7 the gap remains wide: the best upper bound is still $\vartheta(C_7) = 3.3176672\dots$, while the lower bound has been improved four times in the space of two months, each time in the seventh significant digit.

The improvements share a structure. One fixes a base object in $C_7^{\boxtimes 5}$, a set of admissible rules for combining such objects, and a recursion tree describing which rule is applied to which intermediates. Itty, Rosin, Carstensen and Reichman found a 134 753-element independent set in $C_7^{\boxtimes 10}$ [4]; Gao recast it as a binary product lemma over gadgets built from the 367-set of Polak and Schrijver [3] and reached $3.2587891539\dots$ [5]; Buys, Polak and Zuiddam (BPZ) [6, 7] replaced the six numbers of a Gao gadget by seven families with prescribed separations, verified a library of ten combining rules and three terminal codes in Lean, and reached $3.2588279859\dots$; Tandon added a heterogeneous refinement of the binary product and reached $3.2588326203\dots$ [8].

In every one of them the recursion tree is fixed by hand. Tandon is explicit: the ordered split $25 = 6 + 11 + 8$, the rule S_{3b} and the terminal code K_{4a} are kept unchanged from BPZ, and only the representations sitting at the nodes are replaced. He is equally explicit that this is not

optimal, and says so in his concluding section. The rules have been scrutinised; the tree has not been searched. This note takes up that remark and does the search.

The result is a better bound obtained from published data alone, and – more usefully – evidence that the tree is a systematically underexploited parameter rather than a lucky one.

Scope. The improvement is in the seventh significant digit, as were the four before it, and the distance to the upper bound is unaffected: $\vartheta(C_7) - \Theta(C_7)$ is still bounded only by 2.9×10^{-2} , four orders of magnitude larger than everything this line of work has gained in total. Nor is the construction deep. It is one substitution in someone else’s recursion, standing on the heterogeneous refinement that produced the blocks it recombines, and stating it here makes it reproducible in an afternoon by anyone holding the same data; we expect it to be superseded quickly, and by the authors whose data it uses. What we hope survives is the observation that the tree is a free parameter, which applies to whatever construction comes next, and the exhaustive verifications of Section 5, which close doors rather than open them but were expensive to establish and are cheap to reuse.

2 The BPZ framework, briefly

We use the seven-family formulation of BPZ [7] as presented by Tandon [8]. Let $L = \{B, N, A, D, O, H, V\}$ carry the symmetric separation relation $\perp$ given by $B \perp \{O, H, V\}$, $N \perp \{A, D, O, H, V\}$, $A \perp \{N, D, H\}$, $D \perp \{N, A, V\}$, $O \perp \{B, N\}$, $H \perp \{B, N, A\}$, $V \perp \{B, N, D\}$. A *seven-family representation* in a graph G is a collection $F = \{F_\lambda\}_{\lambda \in L}$ of independent sets with F_λ separated from F_μ whenever $\lambda \perp \mu$, and its *cardinality vector* is $(|F_B|, \dots, |F_V|)$ in the order (B, N, A, D, O, H, V) .

An m -ary *combining rule* is a family $T = \{T_\lambda\}$ with $T_\lambda \subseteq L^m$, admissible when any two words of one component are separated in some coordinate and words of components with separated labels are too. For admissible T and representations $F^{(1)}, \dots, F^{(m)}$ the sets $F_\lambda^{\text{out}} = \bigcup_{(\lambda_1, \dots, \lambda_m) \in T_\lambda} F_{\lambda_1}^{(1)} \times \dots \times F_{\lambda_m}^{(m)}$ form a representation in the strong product, with

$$|F_\lambda^{\text{out}}| = \sum_{(\lambda_1, \dots, \lambda_m) \in T_\lambda} \prod_{i=1}^m |F_{\lambda_i}^{(i)}|. \quad (1)$$

A *terminal code* is a set of m -tuples whose Cartesian products are pairwise separated; its union is a single independent set, of size given by the same formula. The BPZ library holds ten combining rules $S_{2a}, S_{2b}, S_{3a}, \dots, S_{3h}$ and three terminal codes K_{3a}, K_{4a}, K_{4b} , all verified admissible in Lean. Gao’s binary product is the rule S_{2a} .

Nothing in (1) requires the inputs to be distinct. This is not a loophole: the published constructions already use it, applying K_{4a} to four copies of one representation.

3 The improvement

Tandon’s Appendix B.4 [8] prints three cardinality vectors, of representations in $C_7^{\boxtimes 30}$, $C_7^{\boxtimes 55}$ and $C_7^{\boxtimes 40}$ respectively:

$$n_6, \quad n_{11}, \quad n_8,$$

and forms $n_{25} = S_{3b}(n_6, n_{11}, n_8)$, a representation in $C_7^{\boxtimes 125}$, then applies K_{4a} to four copies to obtain an independent set in $C_7^{\boxtimes 500}$ of 257 digits, giving 3.258832620353266309...

These three vectors are themselves the output of his heterogeneous recursion, and it is worth being precise about what follows. The construction below introduces no new heterogeneous step, and no propagation of the auxiliary data through one; but it rests entirely on the representations that the heterogeneous refinement produced. It is a better recombination of his objects, not an alternative to them, and without that refinement there would be nothing here to recombine.

Theorem 1. *Let $v = S_{3b}(n_6, n_{11}, n_{11})$ and let $M = K_{4a}(v, v, v, v)$. Then M is the cardinality of an independent set in $C_7^{\boxtimes 560}$, it has 288 digits, and*

$$\Theta(C_7) \geq M^{1/560} = 3.258834362237710794\dots$$

Proof. n_6 and n_{11} are cardinality vectors of seven-family representations in $C_7^{\boxtimes 30}$ and $C_7^{\boxtimes 55}$; S_{3b} is an admissible ternary combining rule and K_{4a} an admissible terminal code, both from the Lean-verified BPZ library. By (1), v is the cardinality vector of a representation in $C_7^{\boxtimes 30+55+55} = C_7^{\boxtimes 140}$, and K_{4a} applied to four copies of it gives an independent set in $C_7^{\boxtimes 560}$. The arithmetic is elementary and exact; the value is recorded in Appendix A. $\square$

The only difference from the published construction is the third input. The split $28 = 6 + 11 + 11$ replaces $25 = 6 + 11 + 8$, and the dimension grows from 500 to 560.

4 The tree is a systematically underexploited parameter

One improvement could be luck. Three observations suggest otherwise.

The published choice is not the best among its own inputs. Drawing three blocks with repetition from $\{n_6, n_8, n_{11}\}$ and the base, in both orientations, under all eight ternary rules and all three terminal codes, and also reading each product directly as an independent set of size $|F_B| + |F_O|$, gives 16 384 candidates. The ranking begins

construction	dimension	bound
$K_{4a}(S_{3b}(n_6, n_{11}, n_{11}))$	560	3.2588343622377
$K_{4a}(S_{3b}(n_8, n_{11}, n_{11}))$	600	3.2588331674194
$K_{4a}(S_{3b}(n_6, n_{11}, n_8))$ (published)	500	3.2588326203533
$K_{4a}(S_{3b}(n_{11}, n_{11}, n_{11}))$	660	3.2588323392139

Only two distinct values stand above the published combination, so it is third by value; among the 16 384 candidates individually it is ninth, because the top value is attained four times. Those four are S_{3b} and S_{3h} , each on n_6 and on its orientation reversal, and they do not merely agree to the printed digits: the four seven-family vectors differ, in the A and D coordinates, yet K_{4a} returns the same 288-digit integer for all of them. The bound is a property of the tree, not of the particular rule or orientation that realises it.

The gain is not monotone. Using the strongest block three times is *worse* than the record. The improvement is a property of the mixture, not of block quality, which is why it is invisible to an argument of the form “use the best available piece”.

Order matters, and by more than the margin being contested. On the published split, permuting the three arguments of S_{3b} costs up to 2.3×10^{-6} – more than half of the 4.6×10^{-6} by which the heterogeneous refinement improved on BPZ. A parameter with that much leverage is worth searching.

Iteration does not compound. Feeding the result back, $X \mapsto S_{3b}(n_6, X, X)$, gains at the first step and loses thereafter: 3.2588343622, then 3.2588110347, then 3.2587875577. The recursion has an interior optimum, so depth alone is not the lever either. Likewise, applying a terminal code to four *distinct* representations was in every case we tried worse than four copies of the best one, which is what one expects of a geometric mean.

5 What does not help

Several natural directions can be closed exhaustively rather than merely explored. We record them because each cost real effort and each is cheap to state.

The base gadget admits exactly eight private pairs. For the 367-set I of Polak and Schrijver [3] in $C_7^{\boxtimes 5}$, a private pair needs a vertex $q \neq r$ with $N[q] \cap I = \{r\}$. Exactly eight members of I own such a vertex, each owning exactly one, and they are precisely the eight pairs in use. So $t = 8$ is a structural ceiling for this set, not an artefact of search. At dimension ten the count is likewise exactly 5152, which is what the recursion already delivers.

The neutral part cannot exceed 322 over the whole orbit. Write U for the closed neighbourhood of the sixteen pair endpoints; then $o = |X| - |X \cap U|$, and U does not depend on the choice of transversals. One more neutral point, $o = 323$, would suffice to beat the record by squaring alone. Applying every automorphism of $C_7^{\boxtimes 5}$ to I while holding X fixed – all $5! \cdot 2^5 \cdot 7^5 = 64\,538\,880$ of them – and imposing the condition $X \cap N[P^H] \cap N[P^V] = \emptyset$, no image reaches $o = 323$. A separate walk of four million size-preserving exchanges on X at fixed $|X| = 367$ also never exceeded 322.

Nine private pairs exist but are unusable. Iterated local search produced 65 distinct 367-sets in $C_7^{\boxtimes 5}$. Their private-pair counts run from $t = 5$ to $t = 10$: two, twenty-one, sixteen, nineteen, six and one respectively. A ninth pair would be worth a great deal: at $s = 367$ any $o \geq 311$ beats the record, and the nine-pair bases offer 315 to 317. They are nonetheless unusable, and the obstruction is quantitative. The doubly covered region $D = N[P^H] \cap N[P^V]$, which the auxiliary set must avoid, grows by almost exactly the same amount with each extra pair. Over all legal splits of every base we hold, its size runs 956 to 988 for the nineteen eight-pair bases – the published X dodges one of size 956 – then 1098 to 1128 for the six with nine pairs, and 1240 to 1268 for the single base with ten. That is about 142 further forbidden vertices per pair, three times over, which makes the obstruction look structural rather than incidental, and closes the ten-pair base for the same reason as the nine-pair ones. Sweeping the whole automorphism group against a nine-pair base – all $64\,538\,880$ images, so the entire orbit, against thirty-two orbit representatives as auxiliary set – no image satisfies the disjointness condition. At $|X| = 366$ the pairs are available but then $o \geq 338$ would be needed, which is out of reach; the question is binary and it answers no.

The published twist is optimal. The heterogeneous step reads $q = |J \setminus N(X^0)|$ for a codebook J , and an automorphic image of an independent set is again one, so the twist is free. Sweeping all 64 538 880 automorphisms, the published choice maximises $q/|J|$ at dimensions 10, 15, 20 and 25 – strictly, at every one of them.

The rule library is saturated where it matters. Admissibility is two local conditions, so one can ask whether the published rules can be extended. No word can be added to the B, N, A, D, H or V component of any rule. The O component is not saturated – S_{3a} extends from four words to seven, S_{3b} from none to four, five rules from none to nine – but this buys nothing, because no terminal code reads the label O , and no component other than O takes O as input. Saturating O and rerunning the published recursion reproduces the published number to the last digit. The three terminal codes admit no extension at all. Randomly generated admissible rules, including four-input rules, which the library does not contain, are far worse than the published ones.

The library is not closed under the orientation involution. Exchanging $A \leftrightarrow D$ and $H \leftrightarrow V$ is an automorphism of the separation relation, so the conjugate of an admissible rule is admissible. Only S_{2a} and S_{3a} are self-conjugate: the conjugates of the other eight rules and of all three terminal codes are admissible but absent. A search that tries both orientations of every input covers them implicitly.

6 Two traps that internal consistency does not catch

Both cost us a day, and both produced numbers that looked like records.

The class rule is a theorem only for homogeneous products. Tracking which vertices lie in $N[X^0]$ under a product is needed to propagate q . For the homogeneous product the rule

$$i(u, w) = (i_u \wedge i_w) \vee (j_u \wedge j_w), \quad j(u, w) = (j_u \wedge i_w) \vee (i_u \wedge j_w)$$

is a three-line theorem: $N[(u, w)] = N[u] \times N[w]$, a rectangle $A \times B$ is met exactly when u meets $N[A]$ and w meets $N[B]$, and X^0 of the product is the union of two rectangles. For the heterogeneous product the neutral part is built from a different rectangle, whose left factor is the escaping part of the codebook, so the rule needs a different predicate E . On the base gadget E differs from j on 5589 of 16807 vertices. The arithmetic of the wrong rule is perfectly self-consistent, and the identity $s = o + h + v$ holds throughout. Recomputing in exact integers does not help: exact arithmetic of a wrong formula stays wrong.

Moreover the predicate E does not close in any bounded class space. Refining the partition of $C_7^{\boxtimes 5}$ by the predicates it generates reaches a fixed point at 16 776 classes among 16 807 vertices; the class degenerates to the vertex. This explains why q_{10} and q_{15} appear in the literature as computed constants rather than as terms of a recursion.

Check against the upper bound. A search that allowed a fifteen-dimensional codebook to serve a ten-dimensional left gadget – an independent set of 49 495 055 vertices in a graph whose independence number is at most 134 753 – produced 4.11, with every internal invariant satisfied. Only comparison with $\vartheta(C_7) = 3.3176672$ exposed it. In a framework where all the arithmetic

is sums of products of non-negative integers, a wrong object does not announce itself; the upper bound is the cheapest independent alarm available.

7 Verification

Everything was computed twice, by implementations sharing no code.

The first is in JavaScript. It parses the rule library from the Lean sources of the BPZ repository at the commit Tandon names, checks that each rule carries the number of words the paper states and that each satisfies both admissibility conditions, and implements (1) in arbitrary-precision integers.

The second is in Python. Its copies of S_{3b} and K_{4a} are typed from the printed tables of Tandon’s Appendix A, not from the machine-readable sources, and it implements the cardinality map independently.

Both are checked against the published construction before being used on ours: each reproduces Tandon’s n_{25} in all seven coordinates and his 257-digit M exactly, and hence his stated bound. On the construction of Theorem 1 they agree on a 288-digit integer and on

$$M^{1/560} = 3.258834362237710794\dots$$

As a sanity check the value lies below $\vartheta(C_7)$, as any lower bound must.

Two further points of external agreement are worth recording, as they were obtained independently of the paper and only afterwards compared with it. Computing the region split of the dimension-ten gadget directly from its explicit vertex sets gives $h = 12\,236$, $v = 16\,744$, which is what Tandon’s Table 6 lists; and $|X_{10} \setminus N(X_{10}^0)| = 134\,689 - 105\,709 = 28\,980$ likewise agrees.

Reproducibility and assistance

The certificate, both verification scripts, the search of Section 4 and the census and obstruction checks of Section 5 are in the accompanying repository, archived at doi:10.5281/zenodo.22727743. The rule library is anchored to commit aa21eeb12b75b0413d3fa9fb4208b5d0bf2c4d65 of the BPZ Lean repository, as in Tandon’s paper.

The search over recursion trees, the exhaustive verifications of Section 5 and the two implementations were carried out with the assistance of a large language model. All results are reproducible from the scripts and certificates supplied; the Python verification was written independently of the search code and from the printed rule tables rather than from the machine-readable sources, so that it does not share the assumptions of the code that produced the result.

References

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A The certificate

The cardinality vector $v = S_{3b}(n_6, n_{11}, n_{11})$ of a seven-family representation in $C_7^{\boxtimes 140}$, in the order (B, N, A, D, O, H, V) . The zero in the O coordinate reflects $T_O^{(3b)} = \emptyset$, as in the published construction.

```

F_B = 630253135055981401870713517244555479224707417188321589099630892251191137
F_N = 144695722184541768107720115813414797839355308000072965335422489005149364
F_A = 234597357441727401048883289845530873859970864445720627213945096783673569
F_D = 279954818401127974982259955129730445935176686603744292186306152889419948
F_O = 0
F_H = 43369967591254229190937078233276375615707839828850482473899658380862816
F_V = 43369967591254229190937078233276375615707839828850482473899658380862816

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Applying K_{4a} to four copies gives an independent set in $C_7^{\boxtimes 560}$ whose cardinality has 288 digits:

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20648129955823938354627223642277723610774954309518268857043903735102296965
39692406790071877020909818803770098048843510624763953396835623197404041433
74478908724579606022957152596859036287499885258150979930066394024808709850
012732304675637453738651340534438147137232028524828404364024830721

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Hence

$$\Theta(C_7) \geq M^{1/560} = 3.258834362237710794\dots,$$

against $3.258832620353266309\dots$ for the published construction. To thirty places the value is $3.258834362237710794353262972358$.